102

Complications of Peritoneal Dialysis

Simon J. Davies, Martin E. Wilkie

CATHETER MALFUNCTION

Optimal Timing and Placement of the Peritoneal Dialysis Catheter

Catheter dysfunction adversely affects patient outcome by preventing commencement of the chosen dialysis modality, as well as by disrupting training schedules and increasing health care costs. Published literature does not demonstrate that one insertion technique is better than another, although a recent meta-analysis suggested an advantage of the laparoscopic compared with the open surgical insertion technique.[1] It is clear that the enthusiasm and experience of the operator are key determinants of catheter outcome,[2] and international guidelines describe the optimal conditions for catheter insertion.[3] Timing is also important; patients randomized to the late start arm of the Initiating Dialysis Early and Late (IDEAL) study (estimated glomerular filtration rate [eGFR] 5–7 mL/min/1.73 m^2) were less likely to start kidney replacement on peritoneal dialysis (PD) than those starting early (eGFR 10–14 mL/min/1.73 m^2), despite PD being their preferred treatment.[4] Early catheter problems are more difficult to manage in the absence of residual kidney function. Our practice is to place catheters once the GFR falls below 10 mL/min/1.73 m^2, aiming to start dialysis when the GFR is 7 to 8 mL/min/1.73 m^2. However, this is individualized according to patient symptoms and whether fluid retention is a particular problem. Some centers use an approach where the external part of the catheter is buried in the subcutaneous tissue at the time of implantation (known as the Moncrief technique) so that it can be exteriorized at the time it is required. This approach means that the catheter can be placed several months prior to be needed. For optimized catheter function, it is necessary that each center audit its success with catheter placement against internationally agreed-upon standards as part of local quality improvement.[2,3]

Catheter Function: Inflow

A 2-L bag of dialysate should normally take 15 minutes or less to run into the peritoneal cavity. If inflow has stopped or significantly slowed, mechanical causes should be suspected. After checking to ensure that the tubing and catheter are not kinked, that all clamps or rollers are open to the inflow position, and that any frangible seal is fully broken, the catheter should be flushed vigorously with 20 mL of heparinized saline. If the catheter is cleared, heparin should be added (500 U/L) to the next few cycles because the cause of the blockage is often a fibrin plug. Should the catheter remain blocked, a plain abdominal radiograph is required. If this shows that the catheter is in a satisfactory position in the deep pelvis, an attempt to restore patency should be made with a thrombolytic agent (urokinase 5000 U or tissue plasminogen activator [tPA] 2 mg in 40 mL of normal saline),[5] which can be instilled into the PD catheter for approximately 1 hour before being withdrawn. If inflow is restored, heparin should be added to the dialysate for the next few cycles. We no longer recommend the use of an endoscopic brush because of safety concerns.

If the radiograph shows the catheter to be malpositioned, an attempt should be made to reposition the catheter tip into the pelvis (Fig. 102.1). This can be done under radiologic guidance with a sterile guidewire inserted into the catheter; alternatively, the catheter can be repositioned at laparotomy or with the laparoscope. Sometimes the catheter becomes wrapped in omentum, suggested usually by complete inflow and outflow failure. This requires a partial omentectomy or an omental hitch, a surgical procedure in which the omentum is temporarily held away from the catheter by a dissolvable suture (omentopexy). The value of laparoscopy in this context is that it can provide a diagnosis as to the cause of catheter flow failure and provide a solution (e.g., by repositioning the catheter, removing an omental wrap, or performing a limited omentectomy).

Catheter Function: Outflow

The most common reason for outflow failure is constipation, although causes of inflow failure discussed previously should also be considered. Loading of the bowel with fecal material is often obvious on a plain radiograph, but treatment for constipation should be initiated without recourse to this investigation because it is so common. Constipation should be treated with oral laxatives or an enema. Subsequently, bowel action should be kept regular by increasing the fiber in the diet and, if necessary, adding a mild laxative. Slow outflow can be a problem in patients using automated peritoneal dialysis (APD), resulting in excessive machine alarms. This can be managed by switching to tidal APD and using a relatively large residual volume (e.g., 25% to 50% of the fill volume). Recently, concerns have been raised about the risk for excessive intraperitoneal dialysate volume that might occur during tidal PD; as a precaution, cycler algorithms have been altered to incorporate a complete drain into the treatment schedule.

Fibrin in the Dialysate

The mesothelial cells of the peritoneal membrane have a range of physiologic functions, including the production of fibrinolytic agents such as tPA. This process is disrupted during peritonitis when the appearance of fibrin in the dialysate is common. If fibrin causes restriction of dialysate flow, heparin (500 U/L) should be added to each bag. A small number of patients have fibrin formation in the absence of peritonitis. Immediately on drainage the bag may appear cloudy, but on standing the fibrin will aggregate and the fluid becomes clear. The first time this happens, a sample must be sent to the microbiology laboratory to exclude infection. If the results of this testing prove negative, the patient can be reassured.

FLUID LEAKS

Fluid leaks occur in which dialysate leaks out of the peritoneal cavity, which can be either visible externally or not. It is recommended that after PD catheter surgery, patients be allowed to heal sufficiently before use (1–2 weeks) to minimize this risk. If the catheter must be used early, low volumes should be used (start with 1 L) in the supine position (e.g., APD with a dry day), with the patient instructed not to mobilize while dialysate is in the peritoneal cavity during the first 2 weeks after catheter insertion. Although PD catheters can be used successfully as the primary approach to manage late-presenting patients or for acute kidney injury, the incidence of leaks is higher under these conditions unless precautions are taken.[6,7]

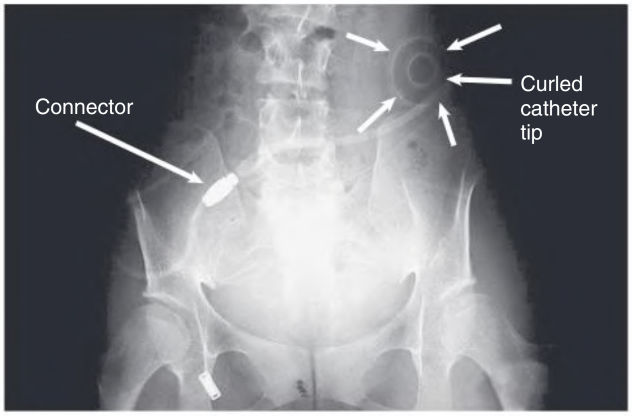

Fig. 102.1 Catheter Malposition. Plain radiograph of the abdomen with curled catheter *(arrows)* misplaced in the upper left abdomen.

External Leaks

On occasion, fluid may leak from the exit site or even the incision used to insert the catheter into the peritoneal cavity. A leak of dialysate, which is confirmed by measuring glucose concentration in the leaking fluid, is a risk factor for infection. It is important that PD catheters be adequately immobilized if used for early-start PD to reduce the risk for tugging and leak.

Internal Leaks

Isolated edema of the abdominal wall suggests an internal leak from the peritoneal cavity, either spontaneously or in association with a surgical hernia. In contrast, genital edema suggests an inguinal hernia or patent processus vaginalis. On occasion, both can be present. The site of the leak can be visualized on computed tomography (CT) scanning after intraperitoneal instillation of contrast material or on magnetic resonance imaging without the use of contrast. It may be necessary for the patient to stand or perform other maneuvers to increase intraabdominal pressure before the leak is demonstrated (Fig. 102.2A). An alternative diagnostic test is to perform scintigraphy after injection of a compound such as technetium 99m–labeled diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA; see Fig. 102.2B). A surgical repair will be required if a major leak is visualized and should always be considered when there is a hernia. Most leaks, however, will heal after resting or with APD, using dry days, or temporary HD.

Hydrothorax

A pleural effusion can occur with generalized fluid overload or local lung disease, but it is occasionally caused by a leakage of dialysate

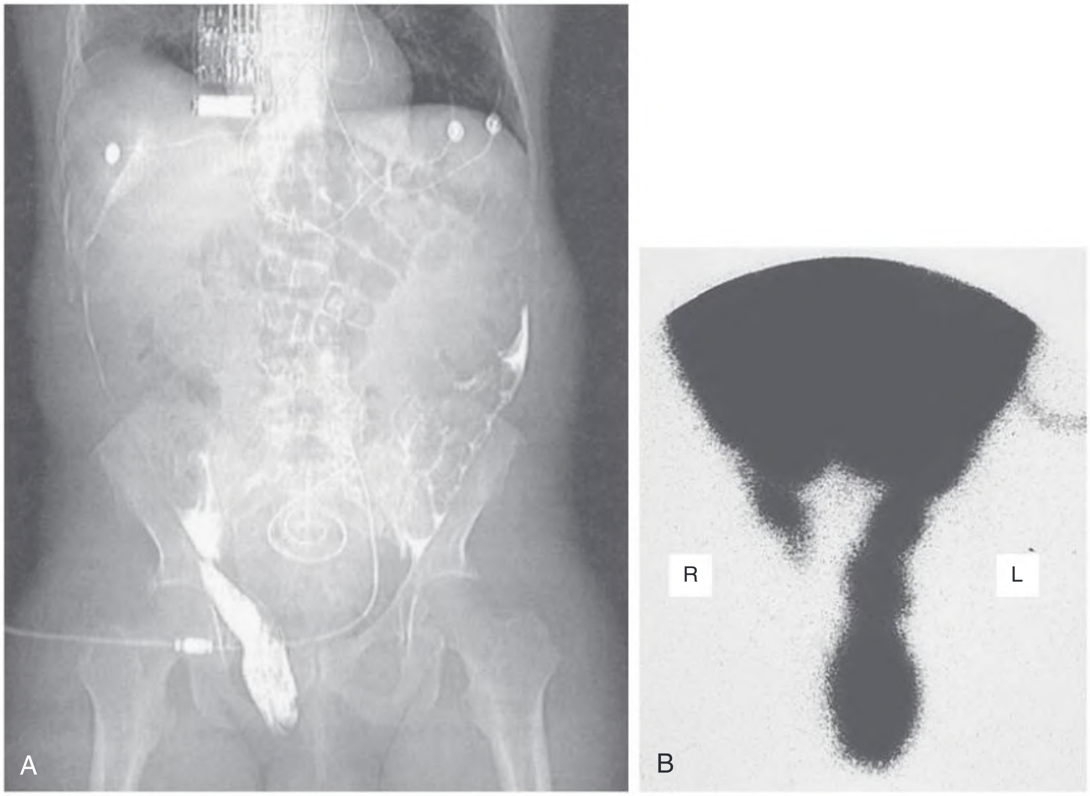

Fig. 102.2 Inguinal Hernia During Peritoneal Dialysis. (A) Computed tomography scan after intraperitoneal injection of contrast material in a male patient showing dialysate flowing into a right inguinal hernia. (B) Peritoneal scintigram of a male patient on peritoneal dialysis showing bilateral inguinal hernias. The left hernia extends into the scrotum; the right hernia is less extensive. (From Tintillier M, Coche E, Malaise J, et al. Peritoneal dialysis and an inguinal hernia. *Lancet.* 2003;362:1893.)

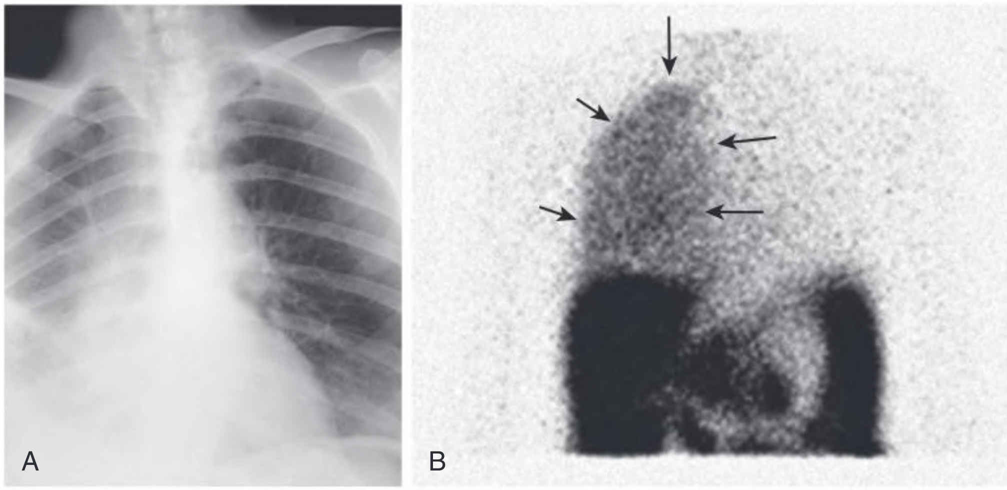

Fig. 102.3 Hydrothorax in Peritoneal Dialysis. (A) Chest radiograph showing a right-sided pleural effusion with partial collapse of the right lung caused by a diaphragmatic leak. (B) Scintigram in a peritoneal dialysis patient showing isotope in the right hemithorax *(arrows)* confirming a right pleural effusion.

through the diaphragm (Fig. 102.3A). This occurs more commonly on the right side. A leak is most simply indicated by aspirating a sample of the effusion and demonstrating that its glucose concentration is higher than the patient's blood glucose concentration, which can be confirmed by scintigraphy after intraperitoneal instillation of isotope, usually ^{99m}Tc-DTPA (see Fig. 102.3B). If one can be confident that the pleural effusion is not caused by the PD, then PD can be continued while the effusion is investigated and managed. Although there are reports that repairing pleural leaks allows subsequent PD, the best advice is to transfer the patient to HD unless there are very strong reasons not to do so.

PAIN RELATED TO PERITONEAL DIALYSIS

Inflow Pain

Soon after starting PD, patients may experience pain during fluid inflow, and occasionally pleuritic pain affects the shoulders, possibly because of diaphragmatic irritation, which usually resolves over the following days. Slowing the rate of fluid inflow will often reduce the symptoms, and peritonitis should be excluded and treated. A small number of individuals have persistent inflow pain, and the use of bicarbonate-lactate–buffered dialysate at physiologic pH improves symptoms in such patients.[8]

Outflow Pain

Some patients have discomfort or pain when the fluid is drained out, which can be experienced in the genital area or rectum and is commonly a result of pelvic irritation related to the catheter tip. This emptying sensation is abolished when the next cycle runs in and is best treated by leaving a small residual volume of fluid in the peritoneal cavity at the end of the drain (e.g., by using tidal APD). In tidal APD, a residual volume is left in the peritoneal cavity at the end of each dialysis cycle (e.g., 20% of the drain volume). To reduce the risk for intraperitoneal volume overload, the treatment algorithm includes at least one complete drain during and at the end of treatment.

Blood-Stained Dialysate

Blood-stained dialysate is uncommon. It is rarely serious but causes considerable alarm to the patient. There is sometimes a clear history of trauma to the abdomen or of unexpected strain. A range of rare conditions are associated with this complication[9]; a few female patients relate the episode to their time of ovulation or menstruation. The treatment is to flush the abdomen with a few cycles of dialysate containing heparin (500 U/L) to minimize the chances of clotting in the catheter. The problem usually resolves spontaneously and often is visible only in one outflow. It is unusual for the blood-stained dialysate to be associated with infection, although it is wise to have the fluid cultured. Routine use of antibiotics is not necessary.

INFECTIOUS COMPLICATIONS

Peritonitis

There are wide variations in peritonitis rates both between and within countries. Reducing peritonitis rates requires a multifaceted, multidisciplinary approach based on the use of preventive measures around the time of catheter insertion, the use of modern disconnect systems, exit site management, and education of patients and health care professionals.[10] This should be supported by regular local audit of peritonitis rates including causative organisms and local sensitivities, which is increasingly important because of the emergence of resistant organisms and the requirement to use antibiotics effectively. Root cause analysis (e.g., inquiring about breaches in sterile technique) should be performed after each episode of PD peritonitis, with retraining as appropriate. Guidelines for the diagnosis and management of PD peritonitis are published by the International Society for Peritoneal Dialysis in adults and children (ISPD; https://www.ispd.org) (Box 102.1).[10,11] The Peritoneal Dialysis Practice and Outcomes Study has provided information on the variation in peritonitis rates between countries and center practices that are associated with lower rates. The findings included advantages for APD use and longer training periods and corroborated the use of antibiotic prophylaxis for catheter insertion and topical application to prevent exit site infection.[12]

Diagnosis of Peritonitis

Peritonitis should be suspected in any patient who develops abdominal pain or a cloudy bag when PD fluid is drained; patients with these symptoms should be advised to contact their dialysis center immediately. Fever may be present but is not a universal feature. Samples of the dialysate should be taken for cell count and microbiologic examination. The diagnosis is confirmed by finding more than 100 white blood cells (WBCs) per microliter (1×10^7/L), which should be more

BOX 102.1 Relevant International Society for Peritoneal Dialysis (ISPD) Guidelines

Cardiovascular and Metabolic

- ISPD Cardiovascular and Metabolic Guidelines in Adult Peritoneal Dialysis Patients
  - Part I: Assessment and Management of Various Cardiovascular Risk Factors
  - Part II: Management of Various Cardiovascular Complications
- Encapsulating Peritoneal Sclerosis
  - Length of Time on Peritoneal Dialysis and Encapsulating Peritoneal Sclerosis: Position Paper for ISPD

Infections

- ISPD Catheter-Related Infections Recommendations: 2017 Update
- ISPD Peritonitis Guideline Recommendations: 2022 Update on Prevention and Treatment

All guidelines available at ISPD.org.

than 50% neutrophils. A Gram stain of the spun deposit should be performed to help identify the type of causative organism, although initial treatment will usually be empiric pending culture and sensitivity results. Various culture techniques have been proposed, but WBC lysis and inoculation into blood culture media is often helpful in increasing the yield of a positive growth.

The dialysate leukocyte count will be affected by dwell length, and this needs to be considered in APD patients. In short dwells, the count will be lower, and under these circumstances, if the proportion of cells that are neutrophils exceeds 50%, empiric treatment of peritonitis should be commenced. Conversely, if the patient has had a dry abdomen during the day, the initial drain on connection may be cloudy. This will clear within one or two cycles, and most of the cells found will be mononuclear leukocytes.

Treatment of Peritonitis

The empiric treatment of peritonitis will vary according to center and should be developed in close collaboration with the local microbiology service, taking into account sensitivity patterns and infection control policy. Initial regimens must cover both gram-positive and gram-negative organisms; the latest ISPD guidelines give examples of appropriate antibiotics and their doses, including vancomycin, cephalosporins, and aminoglycosides. Antibiotic regimens are adjusted as soon as culture results are available, usually after about 48 hours.[10] The preferred route of administration is intraperitoneal (IP) unless the patient has features of systemic sepsis. IP gentamicin can be administered as daily intermittent dosing at a dose of 0.6 mg/kg/day (aiming to maintain the trough serum concentration of <2 mg/L); IP vancomycin also can be administered intermittently every 5 to 7 days, at a dose of 15 to 30 mg/kg, aiming to maintain the serum vancomycin level above 15 µg/mL. IP cephalosporin can be administered either continuously (in each exchange) or on a daily intermittent basis.[10]

Dosage regimens will depend on whether the patient is on continuous APD (CAPD) or APD. For CAPD, the antibiotic is administered as a loading dose in the first bag and then as a maintenance dose in subsequent bags. The intermittent IP dose can be given in the day dwell of APD patients; however, APD results in a higher peritoneal clearance of antibiotics than CAPD, and therefore extrapolation of data from CAPD to APD may result in underdosing.

Once the culture result is available, the regimen should be modified accordingly (Table 102.1). If the organism is methicillin-resistant

TABLE 102.1 Antibiotic Regimens for Bacterial Peritoneal Dialysis Peritonitis

Culture	Antibiotic
Gram positive: based on sensitivities	Vancomycin or other appropriate antibiotic
Coagulase-negative staphylococci	Treat for 14 days
Staphylococcus aureus[a]	Treat for 21 days: screen for carriage
Enterococci	Treat for 21 days
Other streptococci	Treat for 14 days
Gram-negative bacilli or mixed bacterial growth	Continue gram-negative coverage based on sensitivities Consider switching to third- or fourth-generation cephalosporins
Pseudomonas[a] or *Stenotrophomonas* spp.	Two antibiotics based on sensitivity Treat for 21–28 days
Other gram-negative bacilli	Treat for 21 days For mixed gram-negative or gram-negative + gram-positive, consider metronidazole and ampicillin/vancomycin

Suggested antibiotic regimens when dialysate fluid culture is available. Except for culture-negative episodes, empiric treatment is stopped once the sensitivities are known. All antibiotic regimens should be adjusted for local microbiologic practices.

[a]Examine for exit site or catheter tunnel infection.

From Li PK-T, Szeto CC, Piraino B, et al. ISPD peritonitis recommendations: 2016 update on prevention and treatment. *Perit Dial Int.* 2016;36:481–508.

Staphylococcus aureus (MRSA), vancomycin will be continued as part of the regimen. If the culture is negative, empiric therapy should be continued for 2 weeks, assuming there is a clinical response. If a gram-negative organism is identified, the subsequent management will depend on the sensitivity (Fig. 102.4). The isolation of multiple organisms with or without anaerobes strongly suggests perforation of the small or large intestine or biliary system. Metronidazole should be added to the regimen to cover anaerobic organisms, antibiotic therapy augmented to intravenous administration, and prompt surgical review obtained with necessary imaging (CT scan if appropriate).

A wide variety of antibiotics other than those cited have been used with success, and these are documented in the ISPD 2016 guideline.[10] A commonly used strategy is to include an oral quinolone, such as ciprofloxacin. There is debate surrounding the role of aminoglycosides, the advantages being simplicity of use and good coverage for gram-negative organisms; however, there are concerns regarding ototoxicity and nephrotoxicity, the former of which is irreversible. Reports regarding the impact of these agents on residual kidney function are inconsistent, but serious episodes of peritonitis tend to adversely affect residual kidney function. It is advisable to avoid recurrent or protracted courses of aminoglycosides, with an early switch to alternative agents (e.g., third-generation cephalosporins) if they are used as empiric first-line treatment. There is evidence for the use of concomitant *N*-acetylcysteine, which should be considered to prevent the ototoxicity.[13] Current recommendations are that for gram-positive organisms, therapy should be for 14 days except in the case of *S. aureus*, for which 21 days is suggested. For culture-negative episodes, 14 days of therapy should suffice. The same is true in the case of single-organism gram-negative peritonitis.

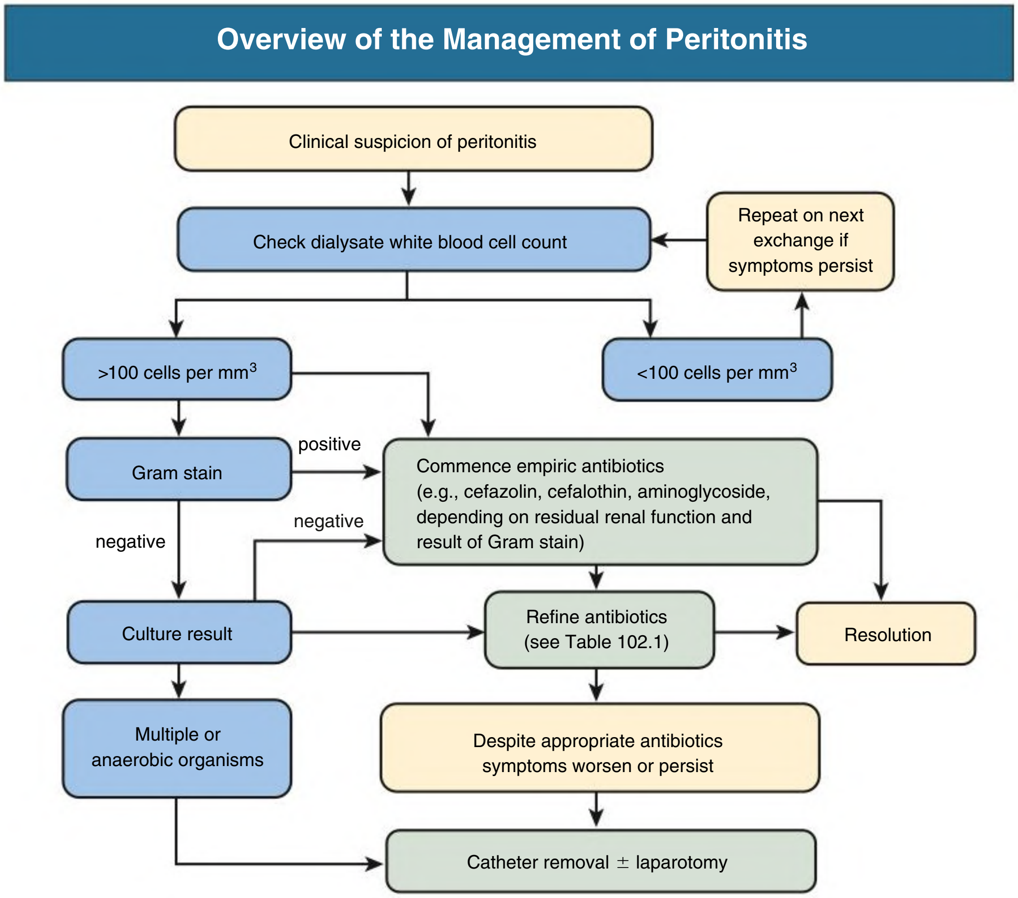

Fig. 102.4 Management of peritonitis.

Many patients can be treated successfully as outpatients. It is extremely important, however, that they be followed either in the clinic or by telephone. In most patients, clinical resolution (as judged by the clearing of the bags) starts within 48 hours. If there is no improvement within 96 hours despite use of the correct antibiotic, as judged by sensitivity tests, the fluid must be retested by cell count, Gram stain, and culture. The ISPD recommends that the catheter be removed if there is no improvement in 5 days; however, in a severe infection this should be done earlier, and in a more indolent case, the period of observation can be longer. A specific recommendation is that the catheter should be removed where a *Pseudomonas* or *S. aureus* peritonitis occurs in conjunction with an exit site or tunnel infection.[10] In addition, the possibility of intraabdominal or gynecologic disease or the presence of unusual organisms such as mycobacteria should be considered. Under these circumstances, a minilaparotomy should be performed to exclude intraabdominal disease, and if mycobacterial infection is suspected, a peritoneal biopsy specimen should be obtained for culture.

Fungal Peritonitis

If peritonitis is caused by yeasts or fungi, the peritoneal catheter should be removed promptly. This should be combined with treatment with an appropriate antifungal for at least 2 weeks after catheter removal. Fungal peritonitis is a serious complication of PD, commonly requiring hospitalization and transfer to hemodialysis (HD), with a high associated mortality. It is essential to review the appropriate antifungal agent with the local microbiologic team; options are described in some detail in the 2016 peritonitis update from the ISPD.[10] Amphotericin B should not be given intraperitoneally because it causes chemical peritonitis and pain. Appropriate antifungal therapy should be continued for at least 2 weeks after catheter removal.

Relapsing Peritonitis

Relapsing peritonitis is defined as infection caused by the same organism as the original infection occurring within 4 weeks, whereas *recurrent peritonitis* is defined as a different organism within 4 weeks of completion of an appropriate course of antibiotics. In general, the advice is to treat as for a primary infection but to try to establish an underlying cause. For example, recurrence of *S. aureus* infection should trigger a search for pericatheter infection. If enterococci or gram-negative organisms are the cause of a relapse, the possibility of intraabdominal disease or an abscess should be considered (although these organisms are frequently waterborne). If a patient has other gastrointestinal symptoms, such as change in bowel habit, appropriate investigation should be conducted. Some organisms (including coagulase-negative staphylococci) produce biofilm that can lead to a relapse of the infection. Consideration should be given to changing the PD catheter once the infection has been treated; of course, the catheter will need to be removed if the infection does not respond to treatment. Current practice in most units is to allow up to 3 weeks of treatment before a new catheter is inserted and to bridge with hemodialysis if required.

Culture-Negative Peritonitis

Culture-negative peritonitis is associated with increased risk for treatment failure. Commonly, the cause relates to the sampling technique or the microbiologic approach; alternatively, concurrent antibiotic use may be responsible. It is important to be aware of the possibility of

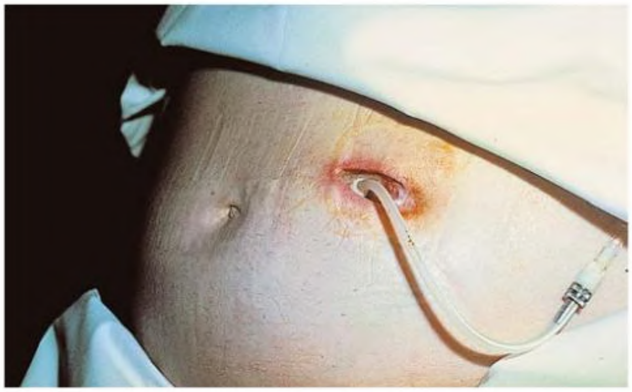

Fig. 102.5 Exit Site Infection. Severe exit site infection that has exposed the outer cuff of the catheter.

fastidious organisms (e.g., atypical mycobacteria, suspicious of contaminated water sources, or yeasts). In addition, other causes of peritoneal inflammation may be responsible; these include the presence of an intraabdominal malignancy or surgical pathology or eosinophilic reactions (e.g., in the case of vancomycin allergy or some fungal infections). An eosinophilic or allergic peritonitis can be diagnosed from the nature of the effluent cell type and requires specific action such as avoiding the causative agent if identifiable, although this usually resolves with time. Chylous effluent is a rare finding in PD; the cause is often unclear, but if recurrent, conditions affecting lymphatic drainage should be considered.

Exit Site Infection

Exit site infection (ESI) is an important complication of long-term PD. The diagnosis is suspected on clinical grounds, usually by the presence of marked erythema or discharge from the exit site (Fig. 102.5). A scoring system for exit sites has been developed to determine the likelihood of infection and grade its severity, with points assigned for crusting, swelling, pain, and discharge according to severity; if the discharge is purulent, this mandates treatment.[14] Extension of the infection into the tunnel may be assessed either clinically or by ultrasound. The most common infecting organism is *S. aureus.* There is now substantial evidence for the use of prophylactic topical antibiotics at the exit site, the strongest being for mupirocin for the prevention of ESI caused by *S. aureus*,[12,14] which we recommend in our practice. There is also evidence for the use of topical gentamicin, although there were some reports of an increase in ESI caused by Enterobacteriaceae, *Pseudomonas* spp., and probably nontuberculous mycobacteria after a change of prophylactic protocol from topical mupirocin to gentamicin.[10]

All suspected infected exit sites should be swabbed; routine swabbing of healthy exit sites should be avoided, and incidental bacterial growth does not require treatment. Unless there is prior evidence that the patient carries MRSA or *Pseudomonas,* initial treatment should be with an antibiotic effective against *S. aureus*—such as a penicillinase-resistant penicillin (e.g., cloxacillin, dicloxacillin, or flucloxacillin; it is important to be aware of the risk for associated hepatotoxicity with the latter) or a first-generation cephalosporin if the patient is allergic to penicillin. In most patients, the drug can be given orally, but if the individual is systemically ill, the antibiotics should be administered intravenously until clinical improvement occurs. Hospitalization, parenteral antibiotics, and often urgent catheter removal are required if there is evidence of spread into the tunnel. If the infection is with MRSA, eradication should be attempted with systemic vancomycin, as for peritonitis. Should the culture grow a gram-negative organism, ciprofloxacin (500–750 mg once daily orally) will be effective empiric treatment in most patients. ESIs caused by *Pseudomonas* spp. are particularly difficult to treat and often require prolonged therapy with two antibiotics.

Treatment is recommended for a minimum of 2 weeks, extended to 3 weeks in *Pseudomonas* infections. In gram-positive infections, if there is no improvement within 7 days, ultrasound of the catheter tunnel should be performed because a collection of fluid around the catheter signifies a tunnel infection. If complete healing does not take place after 4 weeks of therapy, further measures should be considered, such as exteriorizing and shaving the outer cuff, because it may be involved in the infection. If the infection persists or relapses, catheter removal must be considered because there is a high risk that the ESI will lead to peritonitis. It is important that the new exit site be formed in a different part of the anterior abdominal wall.

MEMBRANE DYSFUNCTION AND INSUFFICIENT ULTRAFILTRATION

Recognition and Significance of Membrane Dysfunction

There are two complementary approaches to recognizing membrane dysfunction as the cause of insufficient ultrafiltration (UF). The first is based solely on the net fluid removal (termed *ultrafiltration capacity*) from a standard 4-hour peritoneal equilibration test (PET). The conduct and interpretation of the PET are further discussed in Chapter 101. Account should be taken of the overfill of dialysis bags by the manufacturers, which can be as much as 200 mL. A net UF capacity below 400 mL with a hypertonic exchange (3.86% glucose/4.25% dextrose) is indicative of an inadequate membrane,[15] bearing in mind that this may not be clinically relevant in a patient with well-preserved residual kidney function. If a middle-strength glucose solution is used (2.27%), the equivalent value is 100 mL (again excluding overfill). The main limitation of this approach is that it relies on a single measurement of UF capacity, which is subject to significant error (coefficient of variation is up to 25%) and does not identify the cause. The second approach to recognizing UF failure is more holistic in that it considers patient factors that affect fluid status (e.g., comorbid conditions) and an acceptable glucose exposure required to maintain adequate hydration. Many clinicians now take the view that regular use of hypertonic solutions is not acceptable unless the life expectancy is shorter than the likely time to development of severe injury and its complications, unusual before 5 years.

Insufficient UF is a significant cause of technique failure[16]; it results in low UF, which in turn increases the risk for mortality in anuric patients[17,18] and is also a risk factor for encapsulating peritoneal sclerosis (EPS).[19] Although it is impossible to set a fluid removal goal that applies to all patients, the European Automated Peritoneal Dialysis Outcomes Study (EAPOS) found that anuric patients who did not reach an UF target of more than 750 mL/day had less efficient membranes (specifically lower osmotic conductance; see later) and higher mortality.[18] Patients whose total fluid removal is less than 1 L/day because of poor UF should have their membrane function assessed.

Establishing the Causes of Membrane Dysfunction

Failure of the membrane to ultrafiltrate sufficiently needs to be distinguished from other causes of inadequate peritoneal fluid removal, such as catheter dysfunction, leak, or excessive fluid reabsorption.[14] The latter can occur if the intraperitoneal cavity pressure is too high, suspected if the UF falls after an increase in dialysate volume. If there is doubt, a fall in the dialysate sodium concentration by more than 5 mmol/L at 1 hour into a PET using 3.86% glucose is an indicator of preserved sodium sieving (see later), suggesting that UF is preserved.

There are two main causes of UF failure: a *fast peritoneal solute transfer rate* (PSTR) or a low intrinsic membrane UF efficiency, despite a preserved osmotic gradient, termed *reduced osmotic conductance*. Both can exist at the start of PD or can be acquired with time on therapy, although it is rare for a patient to develop reduced osmotic conductance without also developing a fast PSTR.

Fast PSTR–Related Membrane Dysfunction: Diagnosis and Management

With the 4-hour PET, a dialysate–plasma creatinine ratio greater than average (typically 0.64) could contribute to poor UF. This is because more rapid diffusion of small solute across the membrane leads to earlier dissipation of the osmotic gradient driving UF. Furthermore, once the gradient is lost, membranes with a larger diffusive area will reabsorb fluid more rapidly. A higher than average PSTR is associated with increased mortality, technique failure, and hospitalization,[20,21] whereas this association is ameliorated when APD is used.[20,22] Thus, both theoretically and empirically, the short exchanges used in APD prescription are associated with better outcomes in fast PSTR–associated insufficient UF. Prevention of fluid reabsorption during the long day or night exchange is also required in these patients, and this can be achieved by use of icodextrin (polyglucose solution), which also improves the fluid status[23] and ultrafiltration, reduces episodes of fluid overload and possibly survival.[24,25] The main determinant of fast PSTR is increased intraperitoneal inflammation, which is independent of systemic inflammation.[26] Only the latter is an independent predictor of mortality.

Low Osmotic Conductance–Related Membrane Dysfunction: Diagnosis and Management

This problem should be suspected if UF insufficiency is suspected or there is reduced UF capacity as described earlier. Osmotic conductance is a measure of the efficiency of the peritoneal membrane to ultrafiltrate for a given osmotic agent, typically glucose. Reduced osmotic conductance can be demonstrated quite easily in the clinic. Sodium sieving is the consequence of free water entering the peritoneal cavity via the transcellular endothelial aquaporin pathway, resulting in a drop resulting from dilution in the dialysate sodium concentration; it is independent of the sodium-coupled UF that occurs across endothelial tight junctions. Thus, the absence of a fall in the dialysate sodium concentration 1 hour after using a high glucose exchange indicates the lack of sodium sieving, which implies reduced free water UF. A sodium dip of less than 5 mmol/L is indicative of this type of membrane dysfunction and is associated with worse outcome on PD, including the risk of EPS.[15] The two causes so far identified are reduced aquaporin function, possibly constitutive and thus present at the start of treatment, and progressive fibrosis of the membrane as a result of acquired membrane injury. There is no specific treatment, so clinical effort should focus on prevention (see next section) and timely switch to HD or transplantation.

CHANGES IN PERITONEAL STRUCTURE AND FUNCTION

It is widely assumed that acquired changes in membrane function, a combination of a fast PSTR, and loss of osmotic conductance are related to structural changes in the peritoneal membrane.[27] There is accumulating evidence that continuous exposure to dialysis solution components and repeated episodes of bacterial peritonitis are the main drivers of this process (Fig. 102.6). The relationship between structure and function is becoming clearer: increased PSTR is likely to reflect a greater vascular surface area, whereas loss of osmotic conductance requires an additional mechanism that is associated with progressive membrane fibrosis and an increased risk for EPS. The submesothelial

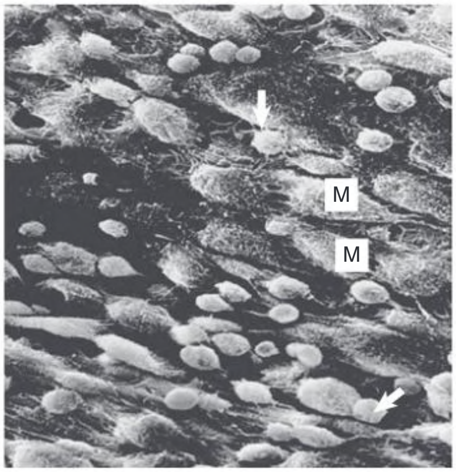

Fig. 102.6 Peritonitis. Scanning electron micrograph of the peritoneum from a patient receiving peritoneal dialysis who has peritonitis. The small round cells *(arrows)* are phagocytes, which are widely distributed among the mesothelial cells *(M)*. (Magnification, ×1800.)

collagenous zone shows progressive increase in thickness with time on PD (Figs. 102.7 and 102.8), and this is associated with progressive changes to the structure of small venules ranging from subtle thickening of the subendothelial matrix to complete obliteration of vessels (Fig. 102.9).[28,29] This process is accompanied by a reduction in sodium sieving, implying reduced free water transport, but no change in aquaporin expression, suggesting that the fibrosis is the main cause of UF failure.[30]

Preventing Membrane Injury

The main clinical factors associated with more rapid and severe membrane injury are early loss of residual kidney function, recurrent or severe peritonitis, and the earlier use of higher glucose–containing solutions (often associated with loss of diuresis but an independent risk factor).[31] Hypertonic solutions may induce injury because of direct glucose toxicity; glucose degradation products (GDPs) manifest as a result of the sterilization process; or both. The prevention of membrane injury should focus on all of these drivers and include (1) preservation of residual kidney function by avoiding both volume depletion and the use of angiotensin-converting enzyme inhibitors and angiotensin receptor blockers; (2) use of loop diuretics to maintain urine volume and delay use of hypertonic exchanges; (3) use of icodextrin in the long exchange; (4) avoidance of peritonitis; and (5) use of low-GDP neutral pH solutions.[32] The balANZ study shows that the increase in PSTR over the first 2 years of PD is prevented by use of ultralow-GDP dialysis fluid. More importantly, this study showed that achievement of less UF, lower use of hypertonic glucose, and in particular low-GDP solutions early in the course of PD, now confirmed in meta-analyses of several trials is associated with preservation of residual kidney function.[24,33] This, combined with trial evidence that fluid status is stable in patients with residual kidney function,[34] supports the main strategy clinicians should adopt in preserving the membrane: prescribing to maintain adequate but not excessive UF, which will only drive thirst and increase the risk for volume depletion.

Encapsulating Peritoneal Sclerosis

A minority of patients on PD develop EPS, in which the bowel is enveloped in a thick cocoon of fibrous tissue, causing intestinal obstruction (Fig. 102.10).[19] It is variable in severity but may be life-threatening, causing death from malnutrition or intraabdominal catastrophe; but with experienced management by a multidisciplinary team, overall

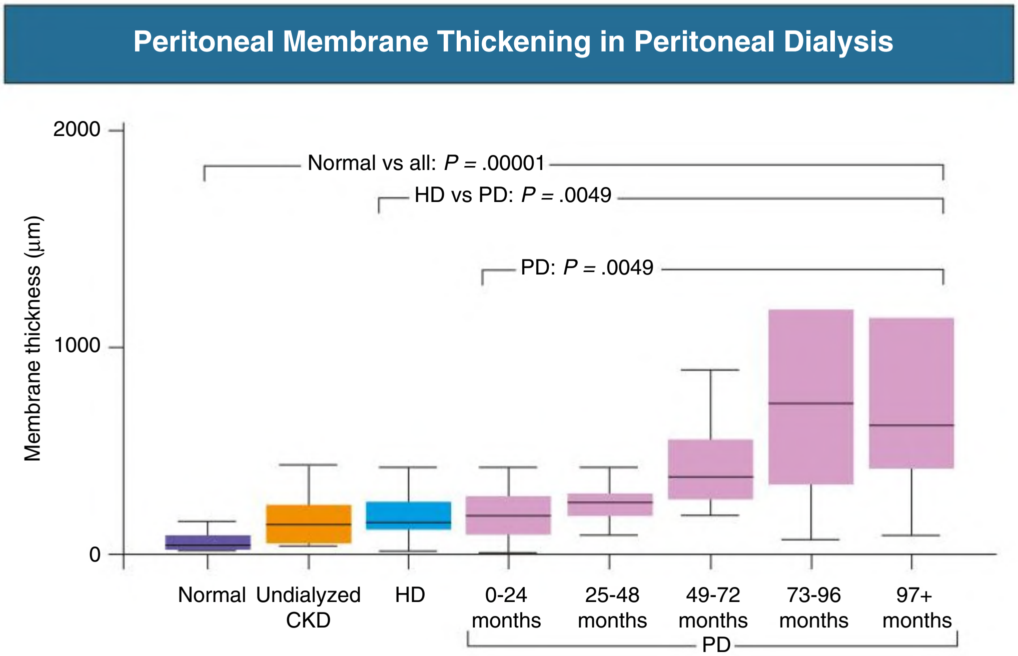

Fig. 102.7 **Peritoneal Membrane Thickening in Peritoneal Dialysis (PD).** The thickness of the submesothelial collagenous zone of the peritoneal membrane in normal individuals, in undialyzed patients with advanced chronic kidney disease (CKD), patients with uremia, in patients receiving hemodialysis (HD), and in those who have received PD for different periods. Membrane thickness is significantly increased in all uremic and dialysis patients compared with normal individuals. Membrane thickness increases significantly with duration of PD and is increased in PD patients as a group compared with HD patients.

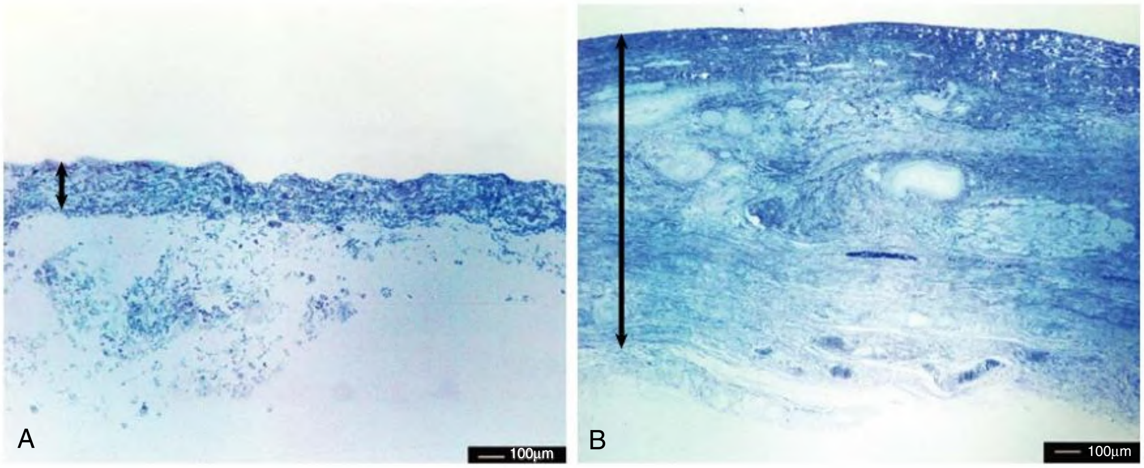

Fig. 102.8 **Morphologic Changes in the Parietal Peritoneal Membrane.** (A) Normal. (B) A patient who has been on peritoneal dialysis for 10 years. Note the marked thickening of the submesothelial compact zone *(arrows)*. (Toluidine blue stain.)

survival compares well with that of matched controls. Diagnosis of this syndrome requires both clinical signs and symptoms of bowel obstruction leading to weight loss and malnutrition (with or without features of systemic inflammation) combined with either typical features on imaging (CT scanning) or confirmation of fibrous cocooning at laparotomy. Although UF failure is a risk factor for EPS (especially when there is also loss of osmotic conductance), EPS appears to be a different pathologic process that predominantly affects the visceral membrane, is usually associated with another trigger (e.g., severe peritonitis or interruption of PD), and frequently has a systemic inflammatory phase (biopsy material more often shows inflammation and fibrinous exudates).

The single most important risk factor for EPS is time on PD; at 5 years the incidence is 2% to 3%, whereas by 10 years, this rises to 6% to 20%.[19] Recent reports from Japan[35] and the Netherlands suggest that the incidence of EPS is decreasing, possibly because of a better understanding of the management of risk factors. There is increasing evidence that surgical treatment (extensive adhesion lysis and excision of the peritoneum while avoiding enterotomy) is most effective, especially when there are obstructive symptoms. This should be undertaken by an experienced surgical team, which can achieve cure rates of 70% to 80%. Parenteral nutrition can be used as a preparation for surgery and occasionally as a long-term solution. In about 50% of patients, symptoms are less severe and gradually resolve. There is no

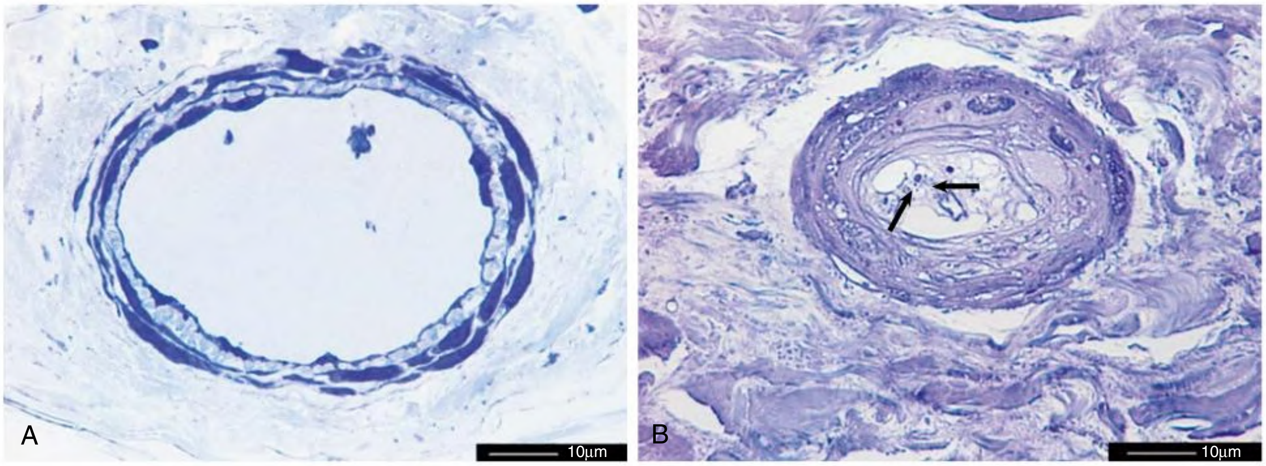

Fig. 102.9 **Blood Vessels in the Parietal Peritoneum: Transverse Sections of Peritoneal Arterioles.** (A) Normal. (B) Vasculopathy in a patient on peritoneal dialysis; the vascular lumen *(arrows)* is occluded by connective tissue containing fine calcific stippling *(arrows)*.

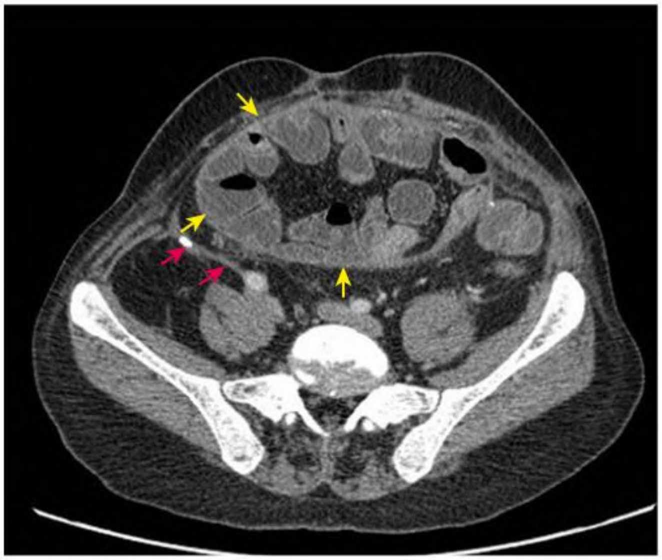

Fig. 102.10 **Encapsulating Peritoneal Sclerosis.** Abdominal computed tomography scan from a patient with encapsulating peritoneal sclerosis. *Red arrows* indicate thickened parietal peritoneum with calcification; *yellow arrows* indicate thickened visceral peritoneum forming a cocoon containing loops of bowel.

role for screening with CT scanning, but this is helpful in diagnosis. Most commonly, EPS develops after transfer from PD to either HD or transplantation; but if it develops in a patient on PD, the consensus is that PD should be stopped to avoid continued exposure to nonphysiologic dialysis solutions. Other strategies, such as continued irrigation, dual-modality treatment with PD and HD, and use of antifibrotic drugs such as tamoxifen, are practiced, but an evidence base for such management is lacking.

NUTRITIONAL AND METABOLIC COMPLICATIONS

Undernutrition

Cross-sectional surveys of patients receiving PD show that about 40% have evidence of mild and 8% severe protein-calorie depletion as judged by the subjective global assessment of nutritional state. Malnutrition is a risk factor for morbidity and mortality of patients on PD and often associated with systemic inflammation. The assessment and management of malnutrition are discussed further in Chapter 90. One obvious contributing factor is protein loss through the peritoneum, which averages 8 g/day. Protein loss is proportional to effective membrane area and so is most marked during peritonitis and in patients with fast peritoneal transport. The ensuing hypoalbuminemia exacerbates the extravascular extracellular fluid expansion in these patients.[36] Therefore, it has been recommended that PD patients should consume daily at least 1.2 to 1.3 g of protein per kilogram of body weight. In practice, many patients take only about 0.8 g/kg/day, and nitrogen balance is maintained partly as a result of calories from dialysate. However, especially once patients are anuric, there is a progressive loss in lean body mass.[34] Despite concerns about the increased prevalence of hypoalbuminemia among peritoneal dialysis patients, the association with mortality is not worse than for HD patients.[37]

PD patients have abnormal eating behavior with smaller meals, slow eating, and impaired gastric emptying, which causes nausea, especially in patients with diabetes. This is worst when using more hypertonic glucose and least severe with icodextrin. Amino acid–based dialysate improves nitrogen balance in malnourished patients, but the long-term nutritional benefit is marginal.[38]

Acid-Base Status

Correction of acidosis is best achieved by use of dialysate with higher levels of potential buffer,[39] but if necessary, oral bicarbonate may be added. There is evidence that correction of acidosis, by whatever means, to within the upper half of the normal range for serum bicarbonate reduces protein catabolism, resulting in weight gain and increased midarm muscle circumference.[38] The use of amino acid–containing dialysate fluid can worsen acid-base status, requiring close monitoring.

Lipids and Obesity

PD results in significant daily glucose absorption, which may range from 80 to 200 g/day (300–800 kcal). Therefore, PD patients tend to develop features of metabolic syndrome, including central obesity, hyperglycemia, dyslipidemia, and hyperinsulinemia; they may even develop frank diabetes, although there is no evidence that this or worsening obesity is more common than in HD patients.[40] These problems can be reduced by use of icodextrin and amino acid solutions in place of glucose with better glycemic control in diabetics.[41] Blood glucose monitoring in diabetic patients using icodextrin must not use glucose dehydrogenase pyrroloquinoline quinone test strips because this will lead to falsely high estimates and risk for severe hypoglycemia.

Despite these concerns, there is no evidence to suggest that PD should be avoided in obese patients even though the survival advantage seen in HD associated with a higher body mass index is not seen with PD.[42]

SELF-ASSESSMENT QUESTIONS

1. Regarding catheter function, which of the following statements are *false?*
   - **A.** Introducing the routine use of laparoscopic technique for catheter insertion will improve early catheter survival more than carefully audited standard methods used by an experienced surgeon.
   - **B.** Slow catheter drainage is most commonly a result of constipation.
   - **C.** Pain on draining in ("inflow pain") is best solved by use of tidal PD.
   - **D.** Pain on draining out is best solved in APD patients by use of tidal PD.
   - **E.** Poor drainage associated with edema in the genital area indicates an inguinal hernia or patent processus vaginalis.
2. Regarding peritonitis and infection, which of the following statements are *false?*
   - **A.** Treatment of suspected peritonitis should always commence with antibiotics covering both gram-positive and gram-negative organisms and antifungal prophylaxis.
   - **B.** Fungal peritonitis usually can be successfully treated with intraperitoneal agents without requiring catheter removal.
   - **C.** Redness and pain at the exit site always should be treated immediately with oral antibiotics.
   - **D.** Prophylactic use of mupirocin or gentamicin at the exit site has been shown to reduce the chances of peritonitis.
   - **E.** Culture of more than one organism in a patient with peritonitis should raise serious concerns of a perforated viscus and lead to minilaparotomy and catheter removal.
3. Regarding ultrafiltration insufficiency, which of the following statements are *false?*
   - **A.** Membrane function should be fully assessed if the ultrafiltration capacity is less than 100 mL on a 4-hour middle-strength glucose 2.27% (dextrose 2.5%) exchange.
   - **B.** There should be a minimum target of 1 L UF per day regardless of residual kidney function.
   - **C.** Fast peritoneal solute transfer rate contributes to poor UF, but this can be addressed by use of APD and icodextrin.
   - **D.** Routine computed tomography scanning to look for progressive membrane thickening and calcification is the best way to screen for EPS.
   - **E.** Progressive loss of osmotic conductance (a 1-hour sodium dip <5 mmol/L using a glucose 3.86% exchange) is thought to represent increasing fibrotic damage, may predispose to EPS, and is a reason for switching to hemodialysis.

REFERENCES

1. Shrestha BM, Shrestha D, Kumar A, Shrestha A, Boyes SA, Wilkie ME. Advanced laparoscopic peritoneal dialysis catheter insertion: systematic review and meta-analysis. *Perit Dial Int*. 2018;38(3):163–171.
2. Goh BL, Ganeshadeva Yudisthra M, Lim TO. Establishing learning curve for Tenckhoff catheter insertion by interventional nephrologist using CUSUM analysis: how many procedures and in which situation? *Semin Dial*. 2009;22:199–203.
3. Crabtree JH, Shrestha BM, Chow KM, et al. Creating and maintaining optimal peritoneal dialysis access in the adult patient: 2019 update. *Perit Dial Int*. 2019;39(5):414–436.
4. Johnson DW, Wong MG, Cooper BA, et al. Effect of timing of dialysis commencement on clinical outcomes of patients with planned initiation of peritoneal dialysis in the ideal trial. *Perit Dial Int*. 2012;32:595–604.
5. Shea M, Hmiel SP, Beck AM. Use of tissue plasminogen activator for thrombolysis in occluded peritoneal dialysis catheters in children. *Adv Perit Dial*. 2001;17:249–252.
6. Povlsen JV, Ivarsen P. How to start the late referred ESRD patient urgently on chronic APD. *Nephrol Dial Transplant*. 2006;21(suppl 2):ii56–ii59.
7. Cullis B, Al-Hwiesh A, Kilonzo K, et al. ISPD guidelines for peritoneal dialysis in acute kidney injury: 2020 update (adults). *Perit Dial Int*. 2021;41(1):15–31.
8. Mactier RA, Sprosen TS, Gokal R, et al. Bicarbonate and bicarbonate/lactate peritoneal dialysis solutions for the treatment of infusion pain. *Kidney Int*. 1998;53:1061–1067.
9. Lew SQ. Hemoperitoneum: bloody peritoneal dialysate in ESRD patients receiving peritoneal dialysis. *Perit Dial Int*. 2007;27:226–233.
10. Li PK-T, Chow MC, Cho Y, et al. ISPD peritonitis guideline recommendations: 2022 update on prevention and treatment. *Perit Dial Int*. 2022;42(2):110–153.
11. Warady BA, Bakkaloglu S, Newland J, et al. Consensus guidelines for the prevention and treatment of catheter-related infections and peritonitis in pediatric patients receiving peritoneal dialysis: 2012 update. *Perit Dial Int*. 2012;32:S32–S86.
12. Perl J, Fuller DS, Bieber BA, et al. Peritoneal dialysis-related infection rates and outcomes: results from the peritoneal dialysis outcomes and practice patterns study (PDOPPS). *Am J Kidney Dis*. 2020;76(1):42–53.
13. Tokgoz B, Ucar C, Kocyigit I, et al. Protective effect of N-acetylcysteine from drug-induced ototoxicity in uraemic patients with CAPD peritonitis. *Nephrol Dial Transplant*. 2011;26:4073–4078.
14. Szeto CC, Li P, Johnson DW, et al. ISPD Catheter-related infections recommendations: 2017 update. *Perit Dial Int*. 2017;37(2):141–154.
15. Morelle J, Stachowska-Pietka J, Öberg C, et al. ISPD recommendations for the evaluation of peritoneal membrane dysfunction in adults: Classification, measurement, interpretation and rationale for intervention. *Perit Dial Int*. 2021. 896860820982218.
16. Davies SJ, Phillips L, Griffiths AM, et al. What really happens to people on long-term peritoneal dialysis? *Kidney Int*. 1998;54:2207–2217.
17. Jansen MA, Termorshuizen F, Korevaar JC, et al. Predictors of survival in anuric peritoneal dialysis patients. *Kidney Int*. 2005;68:1199–1205.
18. Brown EA, Davies SJ, Rutherford P, et al. Survival of functionally anuric patients on automated peritoneal dialysis: the European APD Outcome Study. *J Am Soc Nephrol*. 2003;14:2948–2957.
19. Brown EA, Bargman J, van Biesen W, et al. Length of time on peritoneal dialysis and encapsulating peritoneal sclerosis - position Paper for ISPD: 2017 update. *Perit Dial Int*. 2017;37(4):362–374.
20. Mehrotra R, Ravel V, Streja E, et al. Peritoneal equilibration test and patient outcomes. *Clin J Am Soc Nephrol*. 2015;10:1990–2001.
21. Brimble KS, Walker M, Margetts PJ, et al. Meta-analysis: peritoneal membrane transport, mortality, and technique failure in peritoneal dialysis. *J Am Soc Nephrol*. 2006;17:2591–2598.
22. Johnson DW, Hawley CM, Mcdonald SP, et al. Superior survival of high transporters treated with automated versus continuous ambulatory peritoneal dialysis. *Nephrol Dial Transplant*. 2010;25:1973–1979.
23. Davies SJ, Woodrow G, Donovan K, et al. Icodextrin improves the fluid status of peritoneal dialysis patients: results of a double-blind randomized controlled trial. *J Am Soc Nephrol*. 2003;14:2338–2344.
24. Htay H, Johnson DW, Wiggins KJ, et al. Biocompatible dialysis fluids for peritoneal dialysis. *Cochrane Database Syst Rev*. 2018;10(10):CD007554.
25. Goossen K, Becker M, Marshall MR, et al. Icodextrin versus glucose solutions for the once-daily long dwell in peritoneal dialysis: an enriched systematic review and meta-analysis of randomized controlled trials. *Am J Kidney Dis*. 2020;75(6):830–846.
26. Lambie M, Chess J, Donovan KL, et al. Independent effects of systemic and peritoneal inflammation on peritoneal dialysis survival. *J Am Soc Nephrol*. 2013;24:2071–2080.
27. Davies SJ. Longitudinal relationship between solute transport and ultrafiltration capacity in peritoneal dialysis patients. *Kidney Int*. 2004;66:2437–2445.
28. Williams JD, Craig KJ, Topley N, et al. Morphologic changes in the peritoneal membrane of patients with renal disease. *J Am Soc Nephrol*. 2002;13:470–479.
29. Honda K, Nitta K, Horita S, et al. Morphological changes in the peritoneal vasculature of patients on CAPD with ultrafiltration failure. *Nephron*. 1996;72:171–176.
30. Morelle J, Sow A, Hautem N, et al. Interstitial fibrosis restricts osmotic water transport in encapsulating peritoneal sclerosis. *J Am Soc Nephrol*. 2015;26:2521–2533.
31. Lambie ML, John B, Mushahar L, et al. The peritoneal osmotic conductance is low well before the diagnosis of encapsulating peritoneal sclerosis is made. *Kidney Int*. 2010;78:611–618.
32. Davies SJ, Mushahar L, Yu Z, et al. Determinants of peritoneal membrane function over time. *Semin Nephrol*. 2011;31:172–182.
33. Johnson DW, Brown FG, Clarke M, et al. Effects of biocompatible versus standard fluid on peritoneal dialysis outcomes. *J Am Soc Nephrol*. 2012;23:1097–1107.
34. Tan BK, Yu Z, Fang W, et al. Longitudinal bioimpedance vector plots add little value to fluid management of peritoneal dialysis patients. *Kidney Int*. 2016;89:487–497.
35. Nakayama M, Miyazaki M, Honda K, et al. Encapsulating peritoneal sclerosis in the era of a multi-disciplinary approach based on biocompatible solutions: the NEXT-PD study. *Perit Dial Int*. 2014;34:766–774.
36. John B, Tan BK, Dainty S, et al. Plasma volume, albumin, and fluid status in peritoneal dialysis patients. *Clin J Am Soc Nephrol*. 2010;5:1463–1470.
37. Mehrotra R, Duong U, Jiwakanon S, et al. Serum albumin as a predictor of mortality in peritoneal dialysis: comparisons with hemodialysis. *Am J Kidney Dis*. 2011;58:418–428.
38. Li FK, Chan LYY, Woo JCY, et al. A 3-year, prospective, randomized, controlled study on amino acid dialysate in patients on CAPD. *Am J Kidney Dis*. 2003;42:173–183.
39. Mujais S. Acid-base profile in patients on PD. *Kidney Int Suppl*. 2003:S26–S36.
40. Mehrotra R, Devuyst O, Davies SJ, et al. The current state of peritoneal dialysis. *J Am Soc Nephrol*. 2016;27:3238–3252.
41. Li PKT, Culleton BF, Ariza A, et al. Randomized, controlled trial of glucose-sparing peritoneal dialysis in diabetic patients. *J Am Soc Nephrol*. 2013;24:1889–1900.
42. Ahmadi S-F, Zahmatkesh G, Streja E, et al. Association of body mass index with mortality in peritoneal dialysis patients: a systematic review and meta-analysis. *Perit Dial Int*. 2015;36:315–325.